CSSE 220 – Object-Oriented Software Development

Rose-Hulman Institute of Technology

Lab 16: Robot Roll Call

Name (print): _____ Section: _____

Read the supplied classes in IntelliJ. Record exact output, no output, a compile-time error, or a runtime exception. Predict **before** each run; preserve your original prediction. Uncomment only one trial at a time, run, then comment it again. Omit `End of run` from your answers. Write multiple output lines in order. Explain only results that differ from your prediction.

Reference declarations (unchanged for every trial):

```
Robot robot = new Scout();     Scout scout = new Scout();
Robot medic = new Medic();     Beacon beacon = new Medic();
```

Part A: Predict the output

A1. `robot.signal();`

Predict:

Actual:

Explain if different:

A2. `scout.report();`

Predict:

Actual:

Explain if different:

A3. `scout.signal();`

Predict:

Actual:

Explain if different:

A4. `medic.report();`

Predict:

Actual:

Explain if different:

Part B: Will it compile?

Name: _____

Test B1, B2a, and B2b separately. If the build fails, record the compiler message; do not run an older build. Restore the line to a comment before proceeding.

B1. `beacon.signal();`

Predict:

Actual:

Explain if different:

B2a. `robot.boost();`

Predict:

Actual:

Explain if different:

B2b. `((Medic) scout).signal();`

Predict:

Actual:

Explain if different:

Part C: Casting experiments

For exceptions, record the exception name, not the full stack trace. Each run starts with the original reference declarations on page 1.

C1. `((Scout) robot).boost();`

Predict:

Actual:

Explain if different:

C2. `((Scout) medic).boost();`

Predict:

Actual:

Explain if different:

Submit: Scan both pages into one readable PDF and submit to Lab 16 in Gradescope. Do not submit Java code. B2a and B2b together form question B2.